

Physics-compiled

Louis Haurie

August 30, 2026

Contents

1	Mathematical physics	2
1.1	Topological structures for physics	2
1.1.1	Submanifolds	2
	A. Prerequisites: diffeomorphisms and the inverse function theorem	2
	B. The graph view of submanifolds	3
	C. Level set description of submanifolds	5
	D. Charts and atlases	9
1.1.2	Constructions on submanifolds	10
	A. Vector fields	10
	B. Covectors and the cotangent bundle	12
	C. Metrics and canonical identifications	14
1.1.3	Exterior algebra and cohomology	16
	A. Exterior algebra: definitions	16
	B. Exterior derivative and k-forms	21
	C. Toward cohomology	25
1.1.4	Abstract manifold	26
1.1.5	Summary	27

1 Mathematical physics

1.1 Topological structures for physics

1.1.1 Submanifolds

A. Prerequisites: diffeomorphisms and the inverse function theorem

Before studying submanifolds, we need two tools from multivariable calculus that will be used repeatedly. These results concern open subsets of $\mathbb{R}^m$ and require no notion of manifold.

Let us first start with some definitions and reminders on differentiability. All of this is admitted, but proofs can be found in math textbooks on calculus.

Definition: Differentiability.

Let $U \subset \mathbb{R}^m$ be open and $f : U \rightarrow \mathbb{R}^k$ a map. We say f is **differentiable at** $p \in U$ if there exists a linear map $L : \mathbb{R}^m \rightarrow \mathbb{R}^k$ such that:

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(p+h) - f(p) - Lh\|}{\|h\|} = 0 \quad (1)$$

The linear map L is called the **total derivative** of f at p , and is denoted df_p or $Df(p)$. It is the best linear approximation to f near p .

Remark. This is the correct notion of differentiability for maps between spaces of dimension greater than 1. It is strictly stronger than the existence of partial derivatives: a function can have all partial derivatives at a point without being differentiable here. Differentiability at p implies continuity at p and implies the existence of all directional derivatives, but not conversely.

Definition: The Jacobian matrix. If $f : U \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$ is differentiable at p , its total derivative df_p is a linear map $\mathbb{R}^m \rightarrow \mathbb{R}^k$. In the standard bases of $\mathbb{R}^m$ and $\mathbb{R}^k$, this linear map is represented by the $k \times m$ matrix of partial derivatives:

$$J(p) = \left(\frac{\partial f^i}{\partial x^j}(p) \right)_{\substack{i=1,\dots,k \\ j=1,\dots,m}} = \begin{pmatrix} \frac{\partial f^1}{\partial x^1} & \cdots & \frac{\partial f^1}{\partial x^m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f^k}{\partial x^1} & \cdots & \frac{\partial f^k}{\partial x^m} \end{pmatrix}_p \quad (2)$$

This is called the **Jacobian matrix** of f at p . For a vector $v \in \mathbb{R}^m$, the action of the total derivative is:

$$df_p(v) = J(p)v \quad (3)$$

Remark: chain rule. If $g : V \rightarrow \mathbb{R}^\ell$ is differentiable at $f(p)$, the composition $g \circ f$ is differentiable at p and:

$$d(g \circ f)_p = dg_{f(p)} \circ df_p \quad (4)$$

In terms of Jacobian matrices: $J_{g \circ f}(p) = J_g(f(p)) J_f(p)$. This is the coordinate expression of the chain rule.

Definition: Smoothness. A map $f : U \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$ is called **smooth**, written $f \in \mathcal{C}^\infty(U)$, if all partial derivatives of all orders exist and are continuous on U . In physics, one often only requires as many derivatives as the calculation needs. We will follow the mathematical convention here for precision.

Definition: Diffeomorphism.

Let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^k$ be open sets. A map $\Phi : U \rightarrow V$ is a diffeomorphism if:

- Φ is a bijection,
- Φ is smooth,
- $\Phi^{-1} : V \rightarrow U$ is also smooth.

A diffeomorphism is the natural notion of equivalence in the smooth category: two open sets related by a diffeomorphism are indistinguishable from the point of view of differential calculus.

Remark. A bijective smooth map need not be a diffeomorphism. The map $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3$ is smooth and bijective, but $f^{-1}(y) = y^{1/3}$ is not differentiable at 0. Smoothness of the inverse must be checked separately.

Theorem: Inverse Function Theorem. Let $\Phi : U_p \subset \mathbb{R}^m \rightarrow \mathbb{R}^m$ be smooth and let $p \in U_p$. If $\det J_\Phi(p) \neq 0$, then there exist open sets U'_p and V'_p such that $p \in U'_p$, $\Phi(p) \in V'_p$ and $\Phi|_{U'_p} : U'_p \rightarrow V'_p$ is a diffeomorphism.

In other words: if the best linear approximation to Φ at p (given by the total derivative $d\Phi_p$) is invertible (non-zero determinant of the Jacobian), then Φ itself is locally invertible with smooth inverse.

Proof strategy. The proof proceeds by constructing a contraction mapping $T_y(x) = y + x - \Phi(x)$ whose unique fixed point (guaranteed by the Banach fixed point theorem, using the smallness of $J_\Phi - I_m$ near p) defines $\Phi^{-1}(y)$. Smoothness of the inverse follows from the chain rule applied to $\Phi^{-1} \circ \Phi = \text{id}$. We refer to [1, 5] for the complete proof.

Remark: two other important theorems are corollary of the Inverse Function Theorem:

- **Implicit Function Theorem:** $F : \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r$, $\text{rank } J(p) = r$. Then dF_p is **surjective**, and the level set $F^{-1}(\{c\})$ is locally a graph over $\mathbb{R}^n$.
- **Immersion Theorem:** $f : \mathbb{R}^n \rightarrow \mathbb{R}^{n+r}$, $\text{rank } J(p) = n$. Then df_p is **injective**, and locally f looks like the standard inclusion $\mathbb{R}^n \hookrightarrow \mathbb{R}^{n+r}$.

B. The graph view of submanifolds

Differential geometry studies spaces that may be globally curved or complicated, but which locally resemble ordinary Euclidean space. The central idea is revolving around the concept of manifold: **a set that locally looks like $\mathbb{R}^n$** . As its definition involves some extent of formalism, following the step of ref. [2], we'll start by motivating its usefulness in physics by admitting that the Euclidean space $\mathbb{R}^n$ which is a manifold, and we first define submanifolds of that space.

We start from ordinary multivariable calculus and asks: What properties must subsets of $\mathbb{R}^N$ satisfy to be able to define differential calculus only on it? To answer this question, we will define graphs of smooth functions and use it to get a first understanding of what a submanifold actually is.

Suppose we have a subset M^n of $\mathbb{R}^{n+r}$, where n and r are integers :

$$M^n \subset \mathbb{R}^{n+r} \quad (5)$$

We will call $\mathbb{R}^{n+r}$ the ambient space. M^n is an embedded submanifold of $\mathbb{R}^{n+r}$ if locally near any point $p \in M^n$, one can find r functions $f^a : \mathbb{R}^n \rightarrow \mathbb{R}$ such that :

$$y^a = f_p^a(x^1, \dots, x^n) \quad (6)$$

Where $(x^1, \dots, x^n)$ are n free real coordinates, and $(y^1, \dots, y^r)$ are dependent real coordinates determined by the r smooth functions $(f_p^1, \dots, f_p^r)$.

Definition: graph description of a manifold

More precisely, a submanifold is locally defined around a point p of neighborhood $U_p \subset \mathbb{R}^{n+r}$ if :

$$\exists f_p : \mathbb{R}^n \rightarrow \mathbb{R}^r / M \cap U_p = \{(x, f_p(x)) / x \in \pi(U_p)\} \quad (7)$$

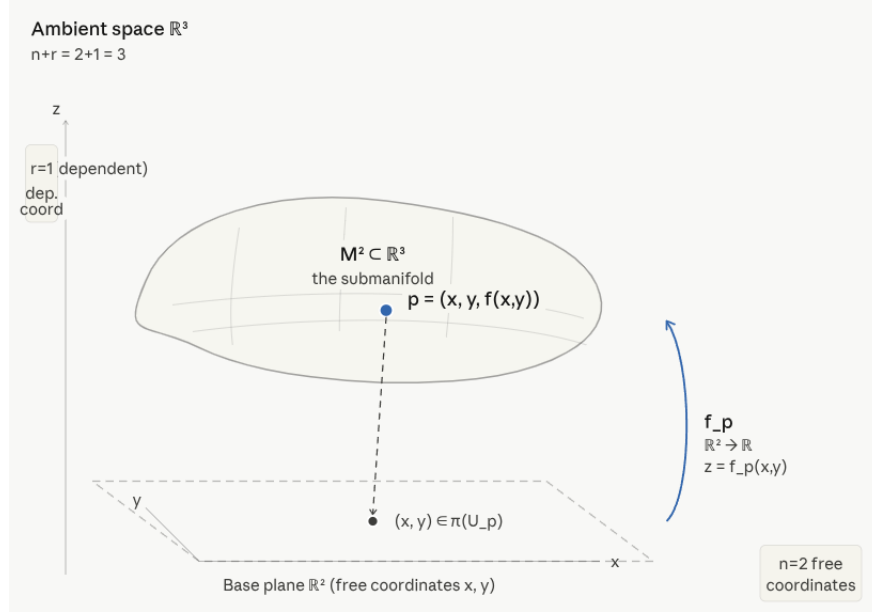


Figure 1: Sketch of a submanifold of $\mathbb{R}^3$: its dimension is 2 (2 free coordinates (x,y)) while the codimension is 1. π is just the projection of a point near p to the n free coordinates, here to the (x,y) plane. Picture generated by A.I., feel free to suggest others if they seem more adapted.

Where $\pi : \mathbb{R}^{n+r} \rightarrow \mathbb{R}^n$ projects onto the n free coordinates. The r smooth maps f_p^a provides a graph representation of the submanifold in the vicinity of one of its point p . n is called the dimension of the submanifold (as these are the free remaining, unconstrained coordinates) while r , the number of constraints or dependent coordinates is called the codimension in the ambient space. The important geometric point to understand is that given $n + r$ independent coordinates, the r maps f_p smoothly constrain r coordinates so that $(x^1, \dots, x^n, y^1, \dots, y^r)$ becomes an n -dimensional family of vectors and remain in the neighborhood of p .

Example: A Curve in $\mathbb{R}^2$. Consider $y = f(x)$. This defines a one-dimensional object inside two-dimensional space. The ambient dimension is 2, but the manifold dimension is 1, because only one variable is free. Once x is chosen, y is fixed. Thus, there is only one degree of freedom. f provides a graph representation of the curve, which turns out to characterize it globally.

Example: A Surface in $\mathbb{R}^3$. Now consider $z = f(x, y)$. This defines a surface in $\mathbb{R}^3$. The ambient dimension is 3, but the manifold dimension is 2, because two variables are free, and the third is constrained. Again, the manifold dimension counts the number of independent local parameters.

A manifold does not have to be globally a graph. The word locally matters. The classic example is the circle: $x^2 + y^2 = 1$. Globally, this cannot be written as: $y = f(x)$, because each value of x corresponds to two possible values of y . However, locally this works perfectly: whether we define it for the upper semicircle $y = +\sqrt{1 - x^2}$ or for the lower semicircle $y = -\sqrt{1 - x^2}$, we can find **local maps** constraining variables on the circle. Note that different local graph representations may be required around different points of the same manifold.

We have so far given sufficient conditions to define the subset M^n as a manifold. Now that we have a rough understanding of what a submanifold is, let's try to characterize it better. To this end, we need to introduce level sets and tangent spaces.

C. Level set description of submanifolds

Definition: constraints

We start by introducing a smooth map over a neighborhood of $p \in \mathbb{R}^{n+r}$ denoted U_p :

$$F_p : U_p \subset \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r. \quad (8)$$

We call F_p the vector of constraints. Each coordinates of F_p are a map $F_p^a : \mathbb{R}^{n+r} \rightarrow \mathbb{R}$ with $a \in \{1, \dots, r\}$ and defines a constraint.

Definition: level set

We introduce :

$$S = F_p^{-1}(\{c\}) = \{x \in \mathbb{R}^{n+r} : F_p(x) = c\} \quad (9)$$

Where c is a vector of $\mathbb{R}^r$. Each of its coordinate is an imposed value for one constraint:

$$x \in F_p^{-1}(\{c\}) \Leftrightarrow \begin{pmatrix} F_p^1(x) \\ \dots \\ F_p^r(x) \end{pmatrix} = \begin{pmatrix} c^1 \\ \dots \\ c^r \end{pmatrix} \quad (10)$$

S is called a level set (or locus in ref. [2]): it is defined as the set of points such that each constraint (coordinates of F_p) satisfy some value (coordinates of c).

Example: The Sphere. Define $F(x, y, z) = x^2 + y^2 + z^2$. Then the sphere is $F^{-1}(\{R^2\}) = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$. Thus, the sphere appears naturally as a constraint surface. As F has only one coordinate, there is only one constraint to this submanifold: it is of dimension 2, codimension 1.

The ambient space $\mathbb{R}^{n+r}$ possesses $n+r$ degrees of freedom. If one imposes r independent constraints $F^a(x) = c^a$, each independent equation removes one degree of freedom. Thus, the remaining number of independent parameters becomes $(n+r) - r = n$. But what does "independent" constraints mean ?

To answer this independency question, we must study the variations of the vector of $\mathbb{R}^r$ they form: F_p . First, let's consider any curve on U_p , the set of points in the neighborhood of p :

$$\begin{aligned} x : \mathbb{R} &\rightarrow U_p \\ t &\mapsto x_p(t) \end{aligned} \quad (11)$$

That are such that $x(0) = p$. The variations of the constraints F_p along this curve is its total derivative:

$$\frac{d}{dt} F_p(x(t))|_{t=0} = dF_p(\dot{x}(0)) = J(p)\dot{x}(0) \quad (12)$$

The partial derivatives of F_p^a are the coordinates of the Jacobian matrix:

$$J^{ij} = \left(\frac{\partial F_p^i}{\partial x^j} \right) = \begin{pmatrix} \frac{\partial F^1}{\partial x^1} & \dots & \frac{\partial F^1}{\partial x^{n+r}} \\ \dots & \dots & \dots \\ \frac{\partial F^r}{\partial x^1} & \dots & \frac{\partial F^r}{\partial x^{n+r}} \end{pmatrix} \quad (13)$$

When we defined the total derivative df_p of a map $f : U \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$, we described it as a linear map $\mathbb{R}^m \rightarrow \mathbb{R}^k$. This is correct but conflates two distinct roles of $\mathbb{R}^m$: as a space of **points** (where

p lives), and as a space of **velocity vectors** at p (where $\dot{x}(0)$ lives). For open subsets of $\mathbb{R}^m$ this distinction is invisible: both are just $\mathbb{R}^m$. But it becomes essential on a submanifold, so we introduce it now.

Definition: tangent space of $\mathbb{R}^m$ at p . Let $p \in \mathbb{R}^m$. The **tangent space** $T_p\mathbb{R}^m$ is the set of all velocity vectors of smooth curves through p :

$$T_p\mathbb{R}^m = \{\dot{x}(0)/x : \mathbb{R} \rightarrow \mathbb{R}^m \text{ smooth, } x(0) = p\} \quad (14)$$

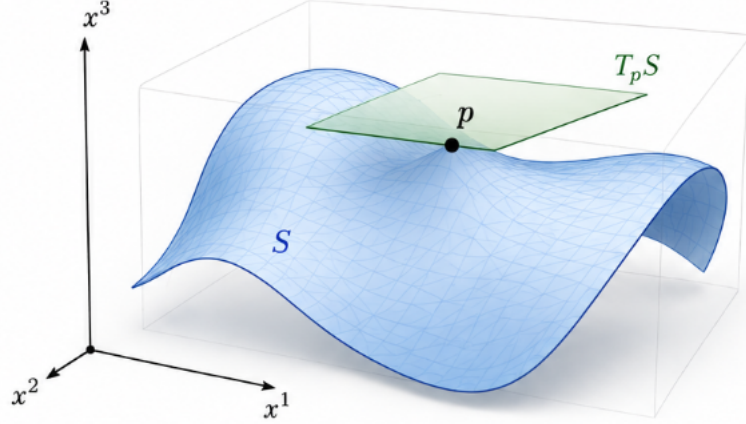


Figure 2: Illustration of the concept of tangent space. Here, the gray box denotes the ambient space, $\mathbb{R}^3$. The blue curve is a submanifold. At $p \in S$, the tangent space sketched as the green plane is a vector space made of all the tangent vector at p . The tangent vectors are obtained by considering the total derivative of trajectories on S crossing p at p . Picture generated by AI, feel free to suggest better illustration for the concept of submanifolds and tangent space.

Since $x(t) = p + tv$ is a valid curve for any $v \in \mathbb{R}^m$, every vector is a tangent vector and $T_p\mathbb{R}^m \cong \mathbb{R}^m$. The tangent space is thus just a copy of $\mathbb{R}^m$ **attached at the point** p , carrying the interpretation of directions rather than positions.

With this language, the total derivative dg_p for a function $g : \mathbb{R}^m \rightarrow \mathbb{R}^h$ is more precisely a linear map:

$$dg_p : T_p\mathbb{R}^m \rightarrow T_{g(p)}\mathbb{R}^h \quad (15)$$

It takes a velocity vector at p in the domain and returns the corresponding velocity vector at $f(p)$ in the target (the velocity of the image curve $f(x(t))$ at $t = 0$). In coordinates, this is still just $J(p)v$, but the interpretation is now geometric: df_p maps directions at p to directions at $f(p)$.

The reason for this language becomes clear on a submanifold $S \subset \mathbb{R}^{n+r}$: a point $p \in S$ still has an ambient tangent space $T_p\mathbb{R}^{n+r} \cong \mathbb{R}^{n+r}$, but not all directions at p stay on S . The tangent space $T_pS \subsetneq T_p\mathbb{R}^{n+r}$ captures only those velocity vectors $\dot{x}(0)$ of curves x that remain on S . It is a strict subspace of dimension $n < n + r$, and will soon show how regular value theorem is precisely what allows identifying it.

Proposition: The tangent space near p of the level set is included in the kernel of the Jacobian matrix:

$$T_pS \subset \ker J(p) \quad (16)$$

Proof

Let us consider a trajectory x defined in eq. 11 near p and belonging to the level set S . Each of its images $x(t) \in U_p \subset \mathbb{R}^{n+r}$ must satisfy $F_p(x(t)) = c$. Let us differentiate :

$$\frac{d}{dt}F_p(x(t)) = \frac{d}{dt}c = 0 \quad (17)$$

Using the matricial representation we introduced with the Jacobian, this equation becomes

$$J\dot{x} = 0 \quad (18)$$

Therefore, any curve in the level set S such that $x(0) = p$ admits a **tangent curve** near p are such that $dF_p(\dot{x}(t)) = 0$, and thus belong in the **kernel of the Jacobian (equivalently, of the linear map dF_p)** at p .

■

We can now clearly answer what independency of constraints means: when does imposing r constraints $F^a(x) = c^a$ actually remove exactly r degrees of freedom? The answer lies in the rank of the Jacobian $J(p)$.

Recall that $J(p)$ is an $r \times (n+r)$ matrix representing the linear map $dF_p : T_p\mathbb{R}^{n+r} \rightarrow T_{F(p)}\mathbb{R}^r$. Its rank is exactly what measures how many of the r constraints are genuinely independent near p .

Definition: Regular point and regular value. A point $p \in S = F^{-1}(\{c\})$ in the level set is called a regular point of F if:

$$\text{rank } J(p) = r \quad (19)$$

That is, if the linear map at p , dF_p , is surjective. The value c is called a regular value of F if every point in the level set $S = F^{-1}(\{c\})$ is regular.

The geometric meaning is immediate: if $\text{rank } J(p) = r$, then the r rows of $J(p)$ are linearly independent. But these rows are exactly $\nabla F^a = (\partial_{x_1} F^a, \dots, \partial_{x_{n+r}} F^a)$, $a \in \{1, \dots, r\}$ meaning they are the derivative of the r constraint functions F^a . Qualitatively, from this results on their variation ∇F^a , the constraints F^a are genuinely pulling in r independent directions in the ambient space at p . Each independent constraint kills exactly one degree of freedom. A more formal way of proving this is invoking the rank-nullity theorem applied to dF_p :

$$\dim \ker J(p) = \dim \mathbb{R}^{n+r} - \text{rank } J(p) = (n+r) - r = n \quad (20)$$

So the kernel has exactly dimension n at a regular point. Since we showed $T_p S \subset \ker J(p)$, **the tangent space at any regular point has at most dimension n** . The inclusion is true for every level set, even singular ones. Equality requires the regularity condition.

Theorem: Implicit function theorem

We already introduced this theorem in the reminders, as a consequence of the inverse function theorem. Here, we state it more rigorously to prove the equivalence between level set and graph views of submanifold.

Consider a point $p \in \mathbb{R}^{n+r}$ of neighborhood U_p and let $F : U_p \subset \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r$ be smooth, $F(p) = c$, and suppose $\text{rank } J(p) = r$. Then, there exist open set $V \subset \mathbb{R}^n$ and $W \subset \mathbb{R}^r$ with $p \in V \times W \subset U$ and a unique smooth map $f : V \rightarrow W$ such that

$$F(x, y) = c \Leftrightarrow y = f(x) \quad \forall x, y \in V \times W \quad (21)$$

In other words, near p , the level set S is exactly the graph of f .

Proof

As $\mathbb{R}^{n+r} \cong \mathbb{R}^n \times \mathbb{R}^r$, we can write $u \in \mathbb{R}^{n+r}$ as $u = (x, y)$ with $x \in \mathbb{R}^n, y \in \mathbb{R}^r$. Introducing $p = (x_0, y_0) \in \mathbb{R}^{n+r}$ of neighborhood U_p , we define the map:

$$\begin{aligned}\Phi : U_p &\rightarrow \mathbb{R}^{n+r} \\ (x, y) &\mapsto (x, F(x, y) - c)\end{aligned}\tag{22}$$

Since $\text{rank} J(p) = r$, the $r \times (n + r)$ Jacobian of F $J(p)$ has r independent columns. The coordinates can be reordered in a way such that the $r \times r$ bottom right $\frac{\partial F}{\partial y}$ block is invertible. Using such reordering, the Jacobian matrix at p of Φ thus is:

$$J_\Phi(p) = \begin{pmatrix} I_n & 0 \\ \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \end{pmatrix}\tag{23}$$

This is a block lower triangular matrix, with $\det J_\Phi(p) = \det \frac{\partial F}{\partial y} \neq 0$. By the inverse function theorem, Φ is a local diffeomorphism near p : there exists a neighborhood U_p of p on which Φ has a smooth inverse Φ^{-1} . In particular if we consider $(x, y) \in \mathbb{R}^{n+r}$ such that $F(x, y) = c$, then $\Phi(x, y) = (x, 0)$ thus $(x, y) = \Phi^{-1}(x, 0)$ and, therefore there exist a smooth function $f : \mathbb{R}^n \rightarrow \mathbb{R}^r$ such that $y = f(x)$. The smoothness of f follows from the fact that $f(x)$ is the second component of $\Phi^{-1}(x, 0)$: since Φ^{-1} is smooth by the inverse function theorem, and projection onto a component is smooth, f is smooth. ■

The rank condition $\text{rank} J(p) = r$ guarantees that the constraints $F^a = c^a$ are non-degenerate at p : the Jacobian pushes the r constrained directions in genuinely independent directions. The implicit function theorem then says: you can always solve locally for r coordinates in terms of the remaining n ; turning the level set view into the graph view. The two descriptions are equivalent near any regular point.

Theorem: Regular Value Theorem.

Let $F : U \subset \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r$ be smooth and c a regular value of F . Then $S = F^{-1}(\{c\})$ **is a smooth submanifold of dimension n** , with tangent space at every $p \in S$:

$$\boxed{T_p S = \ker J(p)}\tag{24}$$

Proof sketch. By the implicit function theorem, near p the level set S is the graph of $f : V \rightarrow W$. A curve on the graph has the form $t \mapsto (x(t), f(x(t)))$, with velocity $(\dot{x}(0), Df(x_0)\dot{x}(0))$ — a vector space of dimension n . Since $\ker J(p)$ also has dimension n by rank-nullity, and $T_p S \subseteq \ker J(p)$, equal finite-dimensional spaces with one included in the other must coincide. The fully explicit construction is in [1]. ■

Remarks

- This theorem is our first sufficient condition to define a smooth submanifold.
- This theorem is local: the implicit function theorem gives a graph representation only in a neighborhood of p , and the regularity condition $\text{rank} J(p) = r$ must hold at each point separately.
- If instead $\text{rank} J(p) < r$ at some point p , the constraints are degenerate there: some constraints are redundant (linearly dependent at p), and the level set may develop a singularity (topologically, we can imagine a corner, a crossing, a cusp... any point where the manifold fails to be locally diffeomorphic to $\mathbb{R}^n$). Such points are called **singular points** and breaks the definition of a smooth submanifold.

Example: the sphere revisited. Once again, consider $F(x, y, z) = x^2 + y^2 + z^2$. From our new knowledge, we consider the associated Jacobian matrix, $J = (2x, 2y, 2z)$. This 1×3 matrix has rank 1 everywhere except at the origin (the null vector does not span any vector space). Since the origin is not on $F^{-1}(\{R^2\})$ for any radius $R \neq 0$, every point of the sphere is regular. The tangent space at $p = (x_0, y_0, z_0)$ is $T_p S^2 = \ker(2x_0, 2y_0, 2z_0) = \{v \in \mathbb{R}^3 : x_0 v^1 + y_0 v^2 + z_0 v^3 = 0\}$. This is exactly the plane tangent to the sphere at p : vectors perpendicular to the radius. The dimension is $3 - 1 = 2$.

Remark: on the choice of which coordinates are "free". The implicit function theorem guarantees that some $r \times r$ submatrix of $J(p)$ is invertible equivalently, that some r coordinates can be expressed in terms of the remaining n . Which ones depends on p and may vary across the manifold. This is why we speak of "free" and "dependent" coordinates only locally. Charts are really the object we use to define which are the free and the dependents coordinate for a submanifold.

D. Charts and atlases

Charts and atlases formalize what "locally looks like $\mathbb{R}^n$ " means. Indeed, the implicit function theorem tells us that near any regular point p , the submanifold S looks like a copy of $\mathbb{R}^n$: we can parametrize it by n free coordinates $(x^1, \dots, x^n)$. This is called a chart.

Definition: Chart

A chart on a submanifold S^n around a point p is a pair (U_p, φ) where:

- $U_p \subset S$ is an open neighborhood of p in S ,
- $\varphi : U_p \rightarrow V_p \subset \mathbb{R}^n$ is a homeomorphism (a continuous bijection with continuous inverse) onto an open set V_p of $\mathbb{R}^n$,
- φ and φ^{-1} are smooth.

The map φ assigns n real numbers $(x^1(q), \dots, x^n(q))$ to each point $q \in U$. These numbers are the local coordinates of q in the chart. The inverse $\varphi^{-1} : V \rightarrow U$ is called a local parametrization of S .

In the graph view, the local parametrization is exactly $\varphi^{-1}(x^1, \dots, x^n) = (x^1, \dots, x^n, f^1(x), \dots, f^r(x))$. The free coordinates x^i are the chart coordinates.

Definition (Atlas)

A collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ such that $\bigcup_\alpha U_\alpha = S$ is called an atlas of S . Every submanifold admits an atlas: the implicit function theorem guarantees a chart around every regular point.

Definition: transition maps

When two charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) overlap (i.e. $U_\alpha \cap U_\beta \neq \emptyset$), we introduce the associated transition map as:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta) \quad (25)$$

Compatibility condition

The compatibility condition assumes that all transition maps are smooth between open subsets of $\mathbb{R}^n$. In other words, the coordinates assigned by two overlapping charts must be smoothly related to each other.

Example: the circle S^1 . Two charts suffice. Let $U_+ = S^1 \setminus \{(0, -1)\}$ (all but the south pole) and $U_- = S^1 \setminus \{(0, 1)\}$ (all but the north pole). On U_+ , use the angle $\theta \in (-\pi, \pi)$; on U_- , use $\theta \in (0, 2\pi)$. These two charts cover S^1 . On the overlap, the transition map is a smooth reparametrization of the angular coordinate. Alternatively and more concretely: the upper and lower semicircle graph representations we defined earlier are essentially two charts. The transition map on their overlap relates $x \mapsto +\sqrt{1-x^2}$ and $x \mapsto -\sqrt{1-x^2}$ near the equator: it is just $x \mapsto x$, which is smooth.

Connection to analytical mechanics

Generalized coordinates are charts. In Lagrangian mechanics, the configuration space of a mechanical system is a manifold Q (e.g. for a double pendulum, $Q = S^1 \times S^1$, a torus). Generalized coordinates $(q^1, \dots, q^n)$ are precisely the coordinates of a chart $\varphi : U \subset Q \rightarrow \mathbb{R}^n$. The Euler-Lagrange equations are written in a particular chart; their form under change of generalized coordinates is the transition map. The fact that physics does not depend on the choice of chart is the coordinate-independence of the Euler-Lagrange equations, which you get for free from the variational formulation.

1.1.2 Constructions on submanifolds

A. Vector fields

To conclude this section we will define an extremely important concept for all physics: a vector field.

Definition: tangent bundle

The collection of all tangent spaces of an n dimensional submanifold S is called the tangent bundle:

$$TS = \bigsqcup_{p \in S} T_p S = \{(p, v) \mid p \in S, v \in T_p S\} \quad (26)$$

This is itself a manifold of dimension $2n$: near each point, a chart on S gives n coordinates for the base point p , and n coordinates for the vector v in the basis induced by that chart.

Definition: vector field.

A vector field on S is a smooth map:

$$\begin{aligned} X : S &\rightarrow TS, \\ p &\mapsto X_p \in T_p S \end{aligned} \quad (27)$$

That assigns to each point p a tangent vector at p . In a chart φ with coordinates $(q^1, \dots, q^n)$, a vector field takes the form:

$$X = X^i(q) \frac{\partial}{\partial q^i} \quad (28)$$

Where the $X^i : V \subset \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth functions. The basis vectors $\frac{\partial}{\partial q^i}$ are **the coordinate vector fields induced by the chart**: the velocity vectors of the coordinate curves $t \mapsto \varphi^{-1}(q^1, \dots, q^i + t, \dots, q^n)$.

In other words, a vector field is a map that "automatically" outputs tangent vector for any point of the submanifold. What is particularly powerful is that it *can* but *does not have to* be induced by charts. In physics, this means it is an object that **does not depend on the coordinate system**.

Smoothness.

X is smooth if and only if for every smooth function $f : S \rightarrow \mathbb{R}$, the function:

$$\begin{aligned} Xf : S &\rightarrow \mathbb{R} \\ (Xf)(p) &= X_p(f) = X^i(p) \frac{\partial f}{\partial q^i}(p) \end{aligned} \quad (29)$$

Is also smooth. This definition is coordinate-independent: it does not depend on which chart you use to compute it.

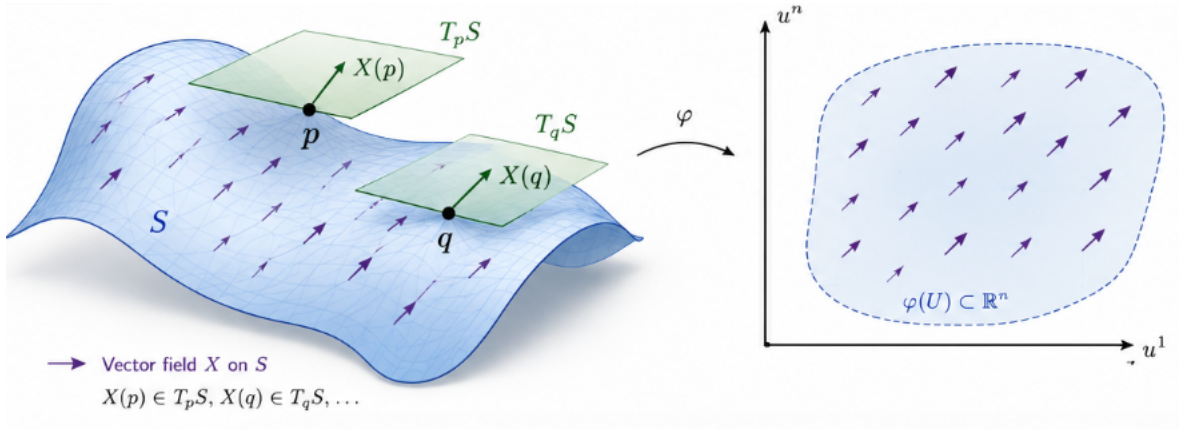


Figure 3: Illustration of a vector field on the submanifold: it is a map that output a vector from the tangent space near the point of evaluation in the submanifold (we showed examples at two points p and q). On the right we apply a chart to define locally the vector field. Picture generated by A.I.: feel free to suggest any picture that may represent better the concept of vector fields.

Vector fields as derivations.

The map $X : \mathcal{C}^\infty(S) \rightarrow \mathcal{C}^\infty(S)$ is linear and satisfies the Leibniz rule:

$$X(fg) = f \cdot Xg + g \cdot Xf \quad (30)$$

This is the intrinsic, coordinate-free characterization of a vector field: it is a derivation on the algebra of smooth functions. On abstract manifolds, this is taken as the definition. This definition is just a generalization to maps of the previously introduced definition. Indeed, since X is defined as a derivation, for a point $p \in S$, $X(f)(p)$ is the total derivative of f at p : $X(f)(p) = df_p(X(p))$. It still is a map that, given a point in the manifold outputs a tangent vector. This time however, that definition considers directly maps on the manifold and output their total derivative (as the vector field satisfies Leibniz rule).

Connection to mechanics

In Newtonian mechanics, Newton 2nd law defines a vector field on the tangent bundle $T\mathcal{S}$ of the state space $\mathcal{S} = (r, \dot{r})$. As Newton equations are defined over the full state space, we can even express the corresponding associated vector field:

$$X_N : \mathcal{S} \rightarrow T\mathcal{S} = (r, \dot{r}) \mapsto (\dot{r}, \frac{1}{m} \sum_k F_k) \quad (31)$$

A curve (called trajectory in this context) $\gamma : \mathbb{R} \rightarrow T\mathcal{S}$ such that

$$\partial_t \gamma = X_N \quad (32)$$

Is a solution of the Newtonian vector field: it solves the equation of motion induced by X_N and is therefore the physical trajectory taken by the system in the regime of validity of Newtonian mechanics.

Similarly, in Lagrangian mechanics, the Euler-Lagrange equations define a vector field on the tangent bundle TQ of the configuration space Q . A solution trajectory $q(t)$ is an integral curve of this vector field: a curve $\gamma : \mathbb{R} \rightarrow TQ$ such that $\dot{\gamma}(t) = X_{\gamma(t)}$ at every t .

The existence and uniqueness of integral curves (at least locally) follows from the Cauchy-Lipschitz theorem applied to the ODE $\dot{\gamma} = X \circ \gamma$. This is the precise sense in which a mechanical system defines a flow on its state space.

B. Covectors and the cotangent bundle

When we defined the tangent space $T_p S$ at a point p of a submanifold S , we obtained a vector space of dimension n : the space of velocity vectors of curves through p . Every vector space has a dual, and the dual of $T_p S$ is equally natural and important.

Definition: covector.

A **covector** at $p \in S$ is a linear map:

$$\xi : T_p S \rightarrow \mathbb{R} \quad (33)$$

The set of all covectors at p is the **cotangent space**:

$$T_p^* S = (T_p S)^* = \{\xi : T_p S \rightarrow \mathbb{R} / \xi \text{ is linear}\} \quad (34)$$

Since $T_p S$ is an n -dimensional vector space, so is $T_p^* S$: they are abstractly isomorphic, but they are *different* spaces with different transformation rules under coordinate changes, as we will see shortly.

Definition: cotangent bundle.

Collecting all cotangent spaces over the submanifold:

$$T^* S = \bigsqcup_{p \in S} T_p^* S = \{(p, \xi) \mid p \in S, \xi \in T_p^* S\} \quad (35)$$

Is known as the **cotangent bundle** of S . Like the tangent bundle, it is a $2n$ -dimensional manifold in its own right. A chart φ on $U \subset S$ induces local coordinates (x^i, ξ_i) on $T^* S|_U$: the first n coordinates x^i locate the base point in S , and the last n coordinates ξ_i give the components of the covector in the dual basis defined hereafter.

Definition: the dual basis and duality.

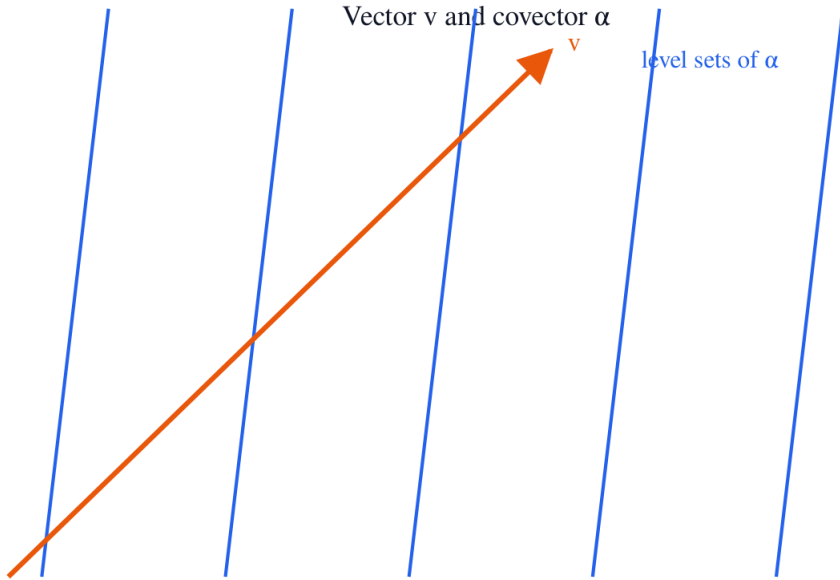


Figure 4: Illustration of what a covector is: the blue line are level sets of the covector α (i.e. lines such that for $v \in T_p S$, $\alpha(v) = \text{cte}$). A covector is therefore a linear form that eats a vector and outputs a scalar, defined at each point p of the manifold S . It lives in the cotangent space at p $T_p^* S$, dual of $T_p S$.

Given a chart φ with coordinates $\varphi(p) = (x^1, \dots, x^n)$ for $p \in S$, the tangent space $T_p S$ has the coordinate basis $\left\{ \partial / \partial x^i \Big|_p \right\}$. The **dual basis** of $T_p^* S$ is the collection of covectors $\{dx^i|_p\}$ forming a basis of $T_p^* S$ and defined by:

$$\boxed{dx^i \Big|_p \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_j^i} \quad \text{DUALITY} \quad (36)$$

The above relationship is called duality and is a very general property between a vector space E and its dual E^* .

Every covector $\xi \in T_p^* S$ expands by definition uniquely in this basis as $\xi = \xi_i dx^i|_p$ for some real coefficients ξ_i , since $\{dx^i|_p\}$ form a basis of $T_p^* S$. Similarly, every tangent vector $v \in T_p S$ expands uniquely as $v = v^i \partial / \partial x^i$. The index placement is deliberate: covector components ξ_i carry *lower* indices, tangent vector components v^i carry *upper* indices, so that $v^i \xi_i$ is a scalar (summed upper and lower index). This operation is called a **contraction**. Note that dx^i is indeed a covector: at $p \in S$, it is a map from $T_p S \rightarrow \mathbb{R}$. With our convention it should technically have a lower index, but despite this we note it using an upper index. This is because x^i carries an upper index (tangent basis induced by a chart) and the d operator preserves index positioning. We will see later what is this d operator. The dual basis *covectors* are *named* after coordinate x^i .

Definition: one-forms.

A smooth section of $T^* S$ (a smooth assignment of a covector to every point) is called a **differential one-form** (or simply a **one-form**). In local coordinates:

$$\omega = \omega_i(x) dx^i \quad (37)$$

where the component functions $\omega_i : V \subset \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth. When evaluated at $p \in \mathcal{S}$, dx^i are the dual basis at $T_p \mathcal{S}$ noted $dx^i|_p$ above. One-forms are basically the dual construction of vector fields:

- A vector field assigns to each point of the manifold a tangent vector at this point.
- A one form assigns to each point of the manifold a covector at this point.

One-forms are the natural objects to integrate along curves: given a curve $\gamma : [a, b] \rightarrow S$, the integral $\int_\gamma \omega = \int_a^b \omega_i(\gamma(t)) \dot{\gamma}^i(t) dt$ is well-defined and independent of parametrization. The most important example: for any smooth function $f : S \rightarrow \mathbb{R}$, the **exterior derivative** df is the one-form:

$$df = \frac{\partial f}{\partial x^i} dx^i \quad (38)$$

whose action on a tangent vector v gives the directional derivative: $df(v) = \frac{\partial f}{\partial x^i} v^i$. In this sense, df is the coordinate-free version of the gradient.

Connection to Hamiltonian mechanics

The cotangent bundle $T^*\mathcal{C}$ of the configuration manifold $\mathcal{C}$ is precisely the **phase space** of a Hamiltonian system. The local coordinates (q^α, p_α) on $T^*\mathcal{C}$ are: q^α for the base point (configuration) and p_α for the covector component (momentum). The generalized momentum $p_\alpha = \partial \mathcal{L} / \partial \dot{q}^\alpha$ is a covector at q because it is defined as a linear form on the tangent space $T_q \mathcal{C}$ (the space of velocities): the map $\dot{q}^\alpha \mapsto p_\alpha \dot{q}^\alpha$ is linear in $\dot{q}^\alpha$ for each fixed q . The one-form $p_\alpha dq^\alpha$ on $T^*\mathcal{C}$ is the **tautological one-form** (or Liouville one-form), and its exterior derivative $\omega = dq^\alpha \wedge dp_\alpha$ is the symplectic form: the fundamental geometric structure of Hamiltonian mechanics. These will be developed further in the analytical mechanics chapter.

C. Metrics and canonical identifications

We have now two bundles over S : the tangent bundle TS (velocity vectors, upper indices) and the cotangent bundle T^*S (linear forms, lower indices). As abstract vector spaces, $T_p S$ and $T_p^* S$ are isomorphic for every p : both have dimension n . But no *canonical* isomorphism exists between them: any identification requires an arbitrary choice of basis, and hence depends on the coordinate system. A **metric** is precisely the additional structure on S that provides a natural, coordinate-independent identification.

Definition: Riemannian metric.

A **Riemannian metric** on S is a smooth assignment, to each point $p \in S$, of an inner product on $T_p S$:

$$\begin{aligned} g_p : T_p S \times T_p S &\rightarrow \mathbb{R} \\ (u, v) &\mapsto g_p(u, v) \end{aligned} \quad (39)$$

that is

- symmetric ($g_p(u, v) = g_p(v, u)$)
- positive and definite ($g_p(v, v) > 0$ for all $v \neq 0$ and $g_p(0, 0) = 0$).

In a local chart with coordinates $(x^1, \dots, x^n)$, the metric is represented by a smooth matrix of functions:

$$g_{\alpha\beta}(x) = g_p\left(\frac{\partial}{\partial x^\alpha}, \frac{\partial}{\partial x^\beta}\right) \quad (40)$$

and the inner product of two tangent vectors $u = u^\alpha \partial_\alpha$, $v = v^\beta \partial_\beta$ is:

$$g_p(u, v) = g_{\alpha\beta}(x) u^\alpha v^\beta \quad (41)$$

The matrix $(g_{\alpha\beta})$ is symmetric and positive definite at every point. A submanifold $S \subset \mathbb{R}^{n+r}$ inherits a natural metric from the ambient Euclidean space by restriction: if $u, v \in T_p S \subset T_p \mathbb{R}^{n+r}$, define $g_p(u, v) = u \cdot v$ (the standard dot product). This is the **induced metric** or **first fundamental form**. With a metric, the usual geometric notions become available on S :

- **Length of a curve** $\gamma : [a, b] \rightarrow S$:

$$\ell(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt = \int_a^b \sqrt{g_{\alpha\beta} \dot{\gamma}^\alpha \dot{\gamma}^\beta} dt \quad (42)$$

- **Angle** between two tangent vectors $u, v \in T_p S$: $\cos \theta = g_p(u, v) / (\|u\| \|v\|)$.
- **Geodesics**: curves of shortest length between two points, satisfying the geodesic equation $\ddot{\gamma}^\alpha + \Gamma_{\beta\gamma}^\alpha \dot{\gamma}^\beta \dot{\gamma}^\gamma = 0$, where $\Gamma_{\beta\gamma}^\alpha$ are the Christoffel symbols built from first derivatives of $g_{\alpha\beta}$.

Definition: musical isomorphisms

Given a metric g , define the map:

$$\flat : T_p S \rightarrow T_p^* S \\ v \mapsto g_p(v, \cdot) \quad (43)$$

In coordinates: $v^\alpha \mapsto \xi_\alpha = g_{\alpha\beta} v^\beta$. This is "lowering an index" (flat map). Since $g_{\alpha\beta}$ is invertible (positive definite implies $\det g \neq 0$), the inverse map exists:

$$\sharp : T_p^* S \rightarrow T_p S \\ \xi \mapsto g^{\alpha\beta} \xi_\beta \quad (44)$$

where $g^{\alpha\beta}$ is the inverse matrix of $g_{\alpha\beta}$. This is "raising an index" (sharp map). Together, $\flat$ and $\sharp$ (the "musical isomorphisms") provide the canonical identification between tangent and cotangent spaces that the metric makes possible.

NB: Why "canonical"?

The identification is canonical in the sense that it does not depend on the choice of coordinates: if you change charts, $g_{\alpha\beta}$ transforms as a covariant tensor (lower indices transform with the forward Jacobian) and v^α transforms contravariantly, so $\xi_\alpha = g_{\alpha\beta} v^\beta$ transforms covariantly: consistently with what a covector should do. The metric absorbs the coordinate dependence of both objects and produces a coordinate-independent pairing.

NB2: And without a metric?

In the absence of a metric, one can still pair a covector $\xi \in T_p^* S$ with a tangent vector $v \in T_p S$ via the natural evaluation $\xi(v) \in \mathbb{R}$, but this is not an identification of the two spaces: it is a map $T_p^* S \times T_p S \rightarrow \mathbb{R}$, not $T_p S \rightarrow T_p^* S$. To go from a tangent vector to a covector without a metric, one needs an additional structure. This is the situation in Hamiltonian mechanics: the Lagrangian provides this structure via the Legendre transform $\dot{q}^\alpha \mapsto p_\alpha = \partial \mathcal{L} / \partial \dot{q}^\alpha$, which plays the role of the metric identification in systems where the kinetic energy defines a natural inner product on TC .

Musical isomorphisms $\flat$ and $\sharp$ (require a metric g)

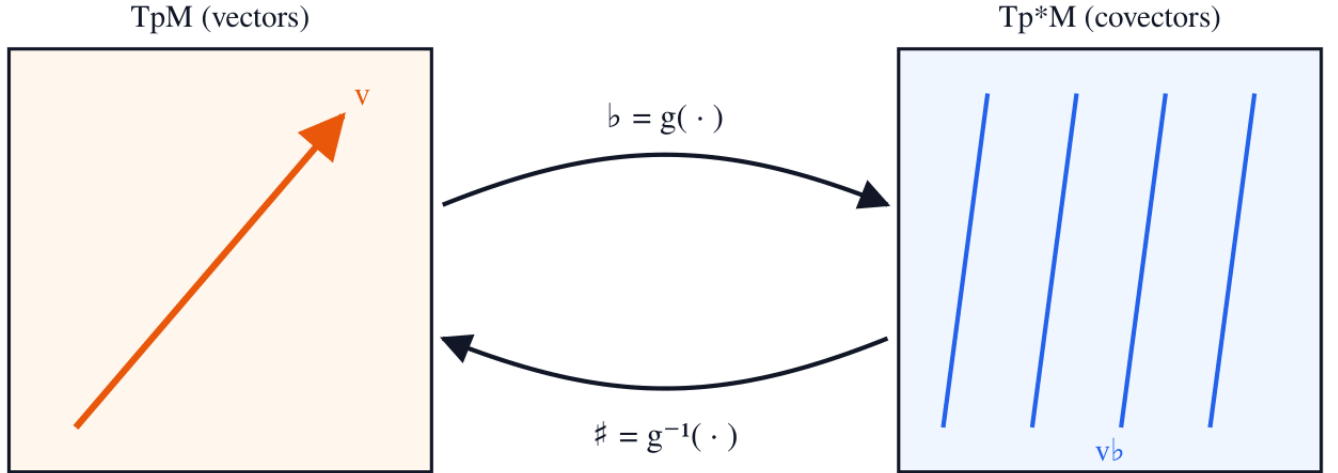


Figure 5: The musical isomorphism $\flat$ (flat) uses the metric to convert a vector $v \in T_p M$ tangent to a manifold M to the dual space ($\sharp(v) = g(v, \cdot)$ is indeed a covector). It is invertible, and the inverse $\sharp$ (sharp) has the opposite action: converting a covector in a vector.

Metrics in physics

Metrics appear throughout theoretical physics in different guises:

- **Kinetic energy metric:** for a mechanical system with Lagrangian $\mathcal{L} = \frac{1}{2}M_{\alpha\beta}\dot{q}^\alpha\dot{q}^\beta - V(q)$, the mass matrix $M_{\alpha\beta}(q)$ is a Riemannian metric on the configuration space $\mathcal{C}$. The Legendre transform $p_\alpha = M_{\alpha\beta}\dot{q}^\beta$ is exactly the $\flat$ map for this metric. Geodesics of $M_{\alpha\beta}$ are the free motions of the system ($V = 0$).
- **Spacetime metric:** in general relativity, the metric $g_{\mu\nu}$ on the 4-dimensional space-time manifold encodes gravity. It is not positive definite (it has Lorentzian signature $(-, +, +, +)$), making it a *pseudo-Riemannian* metric, but the identification between tangent and cotangent spaces works the same way. The geodesics of $g_{\mu\nu}$ are the trajectories of freely falling particles.
- **Fisher information metric:** in statistical mechanics and information theory, the space of probability distributions is itself a manifold, and the Fisher information matrix defines a natural Riemannian metric on it (the *information geometry* of the system). This will appear when we discuss the geometry of statistical states.

1.1.3 Exterior algebra and cohomology

A. Exterior algebra: definitions

We are already familiar with two products between vectors in $\mathbb{R}^3$:

- The **inner product** $\cdot : V \times V \rightarrow \mathbb{K}$, which contracts two vectors belonging to V into a scalar $\mathbb{K}$. It is worth noting that the inner product is a linear symmetric definite positive map. The **dot product** is a special case of inner product, in Euclidian space where the metric is given by $g_{ij} = \delta_{ij}$. Then, the expression of the dot product of two vectors u and v is $u \cdot v = u^i v_i \in \mathbb{R}$

(v is indeed a vector of components v^i , but implicitly transformed into a covector v_i with the Euclidian metric).

- The **cross product** $u \times v \in \mathbb{R}^3$, which produces a new vector perpendicular to both and encoding the oriented area of the parallelogram they span.

Both are special cases of a more general and more powerful construction: the **exterior product**, which generalizes to any dimension, any degree, and any manifold, without requiring a metric. But to define it, we first have to introduce a few definitions.

Definition: tensor products

Given two vector spaces E and F , their tensor product $E \otimes F$ is a new space built by pairing every vector of E with every vector of F . To make sure this new space behaves nicely with linear algebra, we collapse (or force to be equal) any pairs that represent the same linear combinations across both spaces. In coordinates, if $e = e^i \partial_i$ and $f = f^j \partial_j$, then $e \otimes f = e^i f^j \partial_i \otimes \partial_j$ are the components of the tensor in the product basis.

For covectors, the tensor product $\alpha \otimes \beta$ of two one-forms is the bilinear map $(u, v) \mapsto \alpha(u)\beta(v)$. While $(\alpha \otimes \beta)(u, v) = (\beta \otimes \alpha)(v, u)$, the tensor product is not symmetric in general:

$$(\alpha \otimes \beta)(u, v) \neq (\beta \otimes \alpha)(u, v) \quad (45)$$

$\alpha \otimes \beta$ as a grid of coefficients $(\alpha \otimes \beta)_{ij} = \alpha_i \beta_j$

	β_1	β_2	β_3
α_1	$\alpha_1 \beta_1$	$\alpha_1 \beta_2$	$\alpha_1 \beta_3$
α_2	$\alpha_2 \beta_1$	$\alpha_2 \beta_2$	$\alpha_2 \beta_3$
α_3	$\alpha_3 \beta_1$	$\alpha_3 \beta_2$	$\alpha_3 \beta_3$
α_4	$\alpha_4 \beta_1$	$\alpha_4 \beta_2$	$\alpha_4 \beta_3$

Figure 6: Illustration in coordinate of the tensor product of a vector α of a 4-dimensional space and a vector β of a 3-dimensional space. Each cell holds one coefficient: this is a degree 2 array, of dimension 4×3 . The antisymmetric part (Λ^2) is only the set of tensors with $i \neq j$ entries such that $\alpha_i \beta_j = -\alpha_j \beta_i$.

Definition: tensor product spaces

We call the tensor product space of degree k the space generated by tensoring k copies of a vector space. Depending on whether we use the vector space V or its dual V^* , we distinguish two types of pure tensor spaces:

1. **Covariant tensor space** (of degree k), consisting of all k -linear forms on V :

$$T_k(V) = \{f : \underbrace{V \times \cdots \times V}_{k \text{ times}} \rightarrow \mathbb{K} \mid f \text{ is } k\text{-linear}\} = \bigotimes_{i=1}^k V^* \quad (46)$$

2. **Contravariant tensor space** (of degree k), consisting of all k -linear forms on V^* :

$$T^k(V) = \{f : \underbrace{V^* \times \cdots \times V^*}_{k \text{ times}} \rightarrow \mathbb{K} \mid f \text{ is } k\text{-linear}\} = \bigotimes_{i=1}^k V \quad (47)$$

Both are vector spaces of dimension n^k , where $n = \dim(V)$. With this notation, the standard product rule behaves cleanly:

$$T_j(V) \otimes T_k(V) \cong T_{j+k}(V) \quad \text{and} \quad T^j(V) \otimes T^k(V) \cong T^{j+k}(V) \quad (48)$$

Definition: mixed tensor space

A tensor can be partially covariant or contravariant. If we consider r copies of a vector space V and s copies of its dual V^* , we call "mixed tensor space of type (r, s) " the space of tensors r times contravariant and s times covariant:

$$T_s^r(V) = T^r(V) \otimes T_s(V) \quad (49)$$

Mixed tensor spaces are heavily used in physics when introducing metric tensors.

Definition: tensor algebra

The tensor algebra is built from direct sum on the degree of tensor product space:

$$T(V) = \bigoplus_{k=0}^{\infty} T^k(V) \quad (50)$$

With the convention that $T^0(V) = \mathbb{K}$ where $\mathbb{K}$ is the scalar field associated to V .

Definition: The exterior (or "wedge") product.

The exterior product antisymmetrizes the tensor product. It is defined for two vectors α and β by:

$$\alpha \wedge \beta = \binom{p+q}{q} Alt(\alpha \otimes \beta) \quad \alpha \in T^p(V), \beta \in T^q(V) \quad (51)$$

Note the definition works exactly the same for covectors. The binomial coefficient is a normalization convention, while Alt (Alternate) is an antisymmetrizing operation:

$$[Alt(\omega)](v_1, \dots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} sgn(\sigma) \omega(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \quad (52)$$

This is also a good place to note that wedge product $\wedge$ is only defined within a single tensor algebra $T(V)$, unlike the general tensor product $\otimes$ which can freely mixes different spaces. We have, by construction and assuming α is a p -linear form or p -vector and β a q -linear form or q -vector:

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha, \quad \alpha \wedge \alpha = 0 \quad (53)$$

Definition: exterior product space

The wedge product as an oriented parallelogram

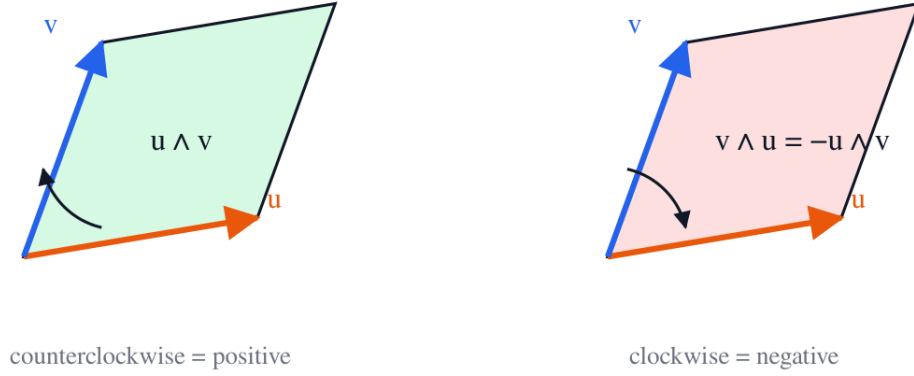


Figure 7: The wedge product of two vectors is called a "bivector" and represents the area of an oriented parallelogram. The antisymmetry is consistent with this picture, as the wedge product of a vector with itself is zero, which is consistent with the fact that the oriented parallelogram formed would have no area.

The *Alt* operator is a projector from the tensor product space of degree k to $\Lambda^k(V) \subset T^k(V)$: this is a subset called exterior product space of degree k that contains only the fully antisymmetrized k -vectors. Note that we can equivalently define $\Lambda_k(V) \subset T_k(V)$ for k linear forms.

Recalling that n is the dimension of V , the dimension of the space of $\Lambda^k(V)$ is:

$$\dim \Lambda^k(V) = \binom{n}{k} \quad (54)$$

As with tensor product spaces, $\Lambda^0 = \mathbb{R}$ (scalars), but here, note that $\Lambda^k = 0$ for $k > n$. The wedge product of two vectors is called a **bivector**, while the wedge product of $k < n$ vectors is a "k-vector" (we also sometimes call "k-blade" k -vectors that can be factored into the product of k independent vectors). We will soon see that the exterior product space has a special meaning for covectors.

Definition: exterior algebra

Finally, we can also define the exterior algebra by analogy to the tensor algebra as:

$$\Lambda(V) = \bigoplus_{k=0}^{\infty} \Lambda^k(V) = \bigoplus_{k=0}^n \Lambda^k(V) \quad (55)$$

We can now give the interpretation of exterior products to covectors and connect it to manifolds. Let us first note that for two covectors $\alpha, \beta \in V^*(= T_1^*(V))$:

$$(\alpha \wedge \beta)(u, v) = \binom{2}{1} \left[\frac{1}{2!} \alpha(u)\beta(v) - \alpha(v)\beta(u) \right] = \alpha(u)\beta(v) - \alpha(v)\beta(u) \quad \forall u, v \in V \quad (56)$$

The result $\alpha \wedge \beta$ is called a **2-form** at p : a bilinear antisymmetric map $V \times V \rightarrow \mathbb{R}$. Geometrically, it measures the signed area of the parallelogram spanned by u and v , projected onto the (α, β) -plane.

The antisymmetry encodes orientation: swapping u and v reverses the sign.

Why can the cross product be identified to vectors?

The cross product in $\mathbb{R}^3$ is a consequence of the special fact that $\dim(\Lambda^2(\mathbb{R}^3)) = \binom{3}{2} = 3$. Thus, bivectors and vectors have the same dimension. However, equal dimension alone is not sufficient to identify these spaces. The identification requires in addition:

- a metric g allowing to identify vector from covector through the induced musical isomorphisms $\sharp$ and $\flat$.
- an orientation, provided by the Hodge star operator $\star : \Lambda^2(\mathbb{R}^3) \rightarrow \Lambda^1(\mathbb{R}^3)$.

Combining these maps gives the chain $\Lambda^2(\mathbb{R}^3) \xrightarrow{\star} \Lambda^1(\mathbb{R}^3) \xrightarrow{\flat} \mathbb{R}^3$, allowing in the special case of $\mathbb{R}^3$ to get $\Lambda^2(\mathbb{R}^3) \simeq \mathbb{R}^3$. In words:

1. Convert the two vectors of $\mathbb{R}^3$ u and v into covectors with the $\flat$ isomorphism.
2. Form their wedge product, this gives a 2 form (the cross product).
3. Apply the Hodge operator, this reduces to a one form.
4. Raise the index (transform the covector back into vectors) using the $\sharp$ isomorphism.

Consequently, for vectors $u, v \in \mathbb{R}^3$, the cross product $\times$ is:

$$u \times v = (\star(u^\flat \wedge v^\flat))^\sharp \quad (57)$$

The cross product is therefore a dimension-3 coincidence, whose identification is made possible with the Hodge star operator. The exterior product is the general operator behind it.

In the dual basis $\{dx^i\}$ of T_p^*S , the k -forms $\{dx^{i_1} \wedge \cdots \wedge dx^{i_k} : i_1 < \cdots < i_k\}$ form a basis of $\Lambda_k(T_p S)$, and a general k -form writes (absorbing in $\omega_{i_1, \dots, i_k}$ the binomial normalization factor of the wedge product):

$$\omega = \frac{1}{k!} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \quad (58)$$

where $\omega_{i_1 \dots i_k}$ is totally antisymmetric in its indices.

Definition: volume form.

The unique (up to scaling) n -form of $\Lambda_n(T_p S)$ with $n = \dim(T_p S)$ is the **volume form**:

$$\text{vol} = dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n \quad (59)$$

It assigns to n tangent vectors the signed volume of the parallelepiped they span. In $\mathbb{R}^3$: $\text{vol}(u, v, w) = \det[u, v, w]$, the scalar triple product. A metric provides a canonical normalization of the volume form.

Definition: interior product

Alongside the exterior product, which raises the degree of a form; we can introduce the interior product (or contraction) ι_X which lowers it. Given a vector field X and a k -form $\omega \in \Lambda_k(V)$, $\iota_X \omega \in \Lambda_{k-1}(V)$ is defined by inserting X into the first slot:

$$[\iota_X(\omega)](v_1, \dots, v_{k-1}) = \omega(X, v_1, \dots, v_{k-1}) \quad (60)$$

Alternatively, the interior product applied on a k -vector will, given a one form α , lead to a $(k - 1)$ -vector as:

$$\iota_\alpha(v_1 \wedge \dots v_k) = \sum_{i=1}^k (-1)^i \alpha(v_i) v_1 \bigwedge_{j \neq i}^k v_j \quad (61)$$

For a one-form α , the interior product reduces to a simple evaluation: $\iota_X \alpha = \alpha(X)$. No metric is required, this is a purely linear-algebraic operation.

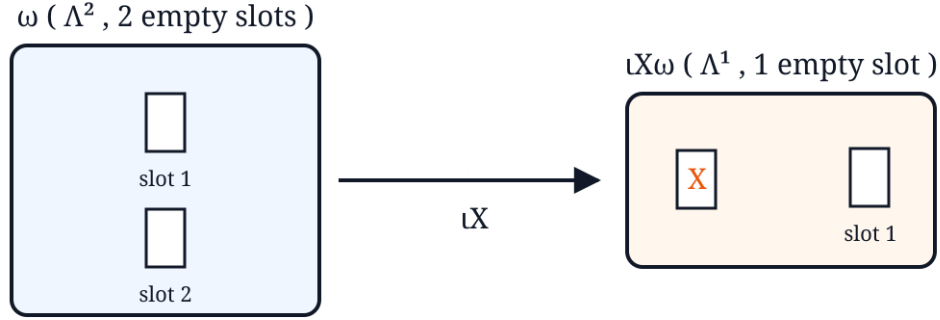


Figure 8: Illustration of the effect of the interior product on a 2-form. It "plugs" a vector field in the first slot of a two-forms, resulting in a one-form. Alternatively, it plugs a one-form to a bivector, resulting in a vector. An interior product is just evaluating a slot with the dual construction.

Beware of the term "outer product", as it is sometimes used to mean "tensor product" and sometimes to mean "exterior product".

B. Exterior derivative and k-forms

Why does a single operator deserve to replace gradient, curl, and divergence? In elementary vector calculus these are three unrelated-looking constructions, each with its own formula, and curl exists only in $\mathbb{R}^3$.

The exterior derivative d replaces all three with one operator, defined on forms of any degree, on a manifold of any dimension, requiring no metric whatsoever. Gradient, curl, and divergence are nothing more than d , viewed through the metric-dependent lens of $\flat$, $\sharp$ and the Hodge star $\star$. This is precisely why d survives unchanged into settings with no fixed metric, or an indefinite one (general relativity), while "curl" as a vector operation does not.

In the following, we consider back $T_p S$ as the tangent space of an n -dimensional submanifold S .

We already introduced the differential df of a smooth function $f : S \rightarrow \mathbb{R}$ as the one-form $df = \frac{\partial f}{\partial x^i} dx^i$. The **exterior derivative** d extends this operation to forms of all degrees, in a way that is intrinsic (coordinate-free) and satisfies one remarkable identity.

Definition: exterior derivative.

The exterior derivative is the unique family of linear maps $d : \Lambda_k(TS) \rightarrow \Lambda_{k+1}(TS)$, for all $k \geq 0$, satisfying:

1. **On functions:** $df = \frac{\partial f}{\partial x^i} dx^i$ (the differential, already defined).
2. **Linearity:** $d(\alpha + \beta) = d\alpha + d\beta$.

Exterior derivative: curl (2-form) and divergence (3-form)

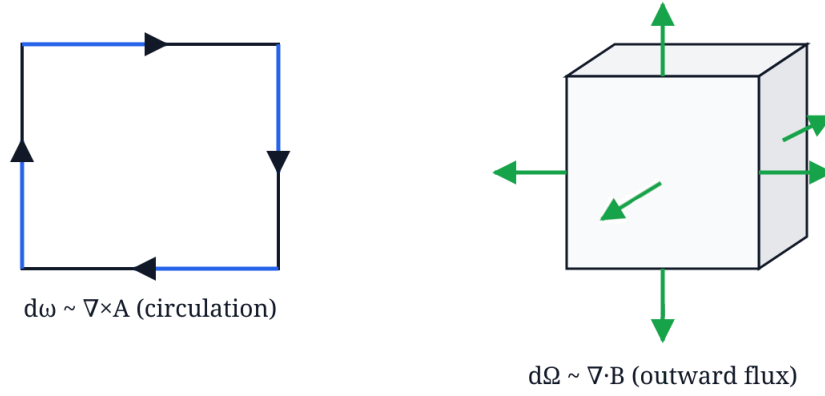


Figure 9: Illustration of how two and three forms are directly related geometrically to curls and divergences.

3. **Graded Leibniz rule:** for a k -form α and any form β :

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta \quad (62)$$

4. **Nilpotency:** $d \circ d = 0$, i.e. $d^2\omega = 0$ for any form ω .

These four properties uniquely determine d in a coordinate independent way. In local coordinates, the action on a general k -form (58) is:

$$d\omega = \frac{\partial \omega_{i_1 \dots i_k}}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \quad (63)$$

Proposition: Two-forms and the curl.

For a one-form $\omega = \omega_i dx^i$, the exterior derivative is the 2-form:

$$d\omega = \frac{\partial \omega_i}{\partial x^j} dx^j \wedge dx^i = \sum_{i < j} \left(\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} \right) dx^i \wedge dx^j \quad (64)$$

The antisymmetric combination $\frac{\partial \omega_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j}$ is the **curl** of ω . In $\mathbb{R}^3$, identifying the one-form $\omega = A_x dx + A_y dy + A_z dz$ with the vector field $\vec{A}$, the 2-form $d\omega$ corresponds to $\nabla \times \vec{A}$ (the standard curl of vector calculus). The condition $d\omega = 0$ (a closed form) says the curl vanishes.

Proposition: three-forms and the divergence.

For a two-form $\Omega = \Omega_{ij} dx^i \wedge dx^j$ in $\mathbb{R}^3$, $d\Omega$ is a 3-form proportional to the divergence of the associated vector field. Explicitly, identifying the 2-form $\Omega = B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy$ with $\vec{B}$:

$$d\Omega = \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) dx \wedge dy \wedge dz = (\nabla \cdot \vec{B}) \text{vol} \quad (65)$$

Therefore, exterior algebra and derivative is a natural tool one can use to write many physical equations.

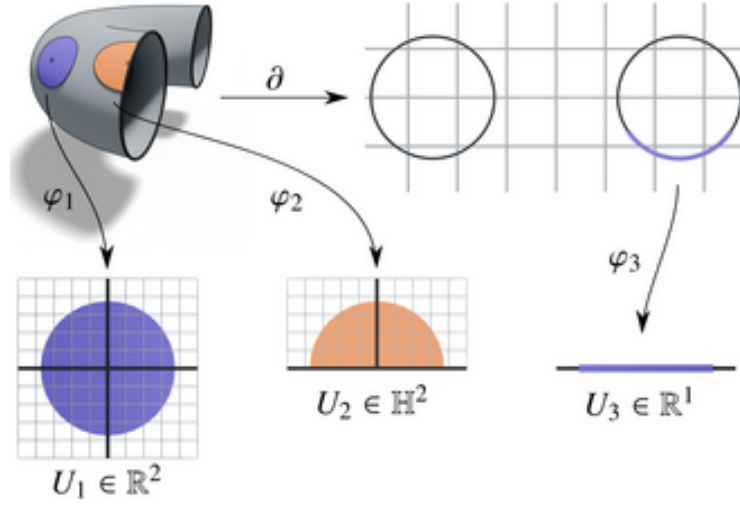


Figure 10: Example of a 2-dimensional submanifold of $\mathbb{R}^3$. The edges of the bent cylinder $\mathcal{C}$ are two circles $\partial\mathcal{C}$. Indeed, we can fit a ball (disk in 2d) in the vicinity of any points beside the one sitting on the two circles, as shown by φ_1 (chart of an interior point) and φ_2 (chart of an edge point). $\partial\mathcal{C}$ is a 1 dimensional submanifold itself: every point on it is an interior point (a ball in 1d is simply an interval $[-\epsilon, \epsilon]$, and it is possible to find such interval for any point on the circle). Therefore: $\partial(\partial\mathcal{C}) = \emptyset$. Figure taken from [Wikipedia](#).

But we can do more than a mere reformulation of the standard way of writing physics equation in $\mathbb{R}^3$. By far one of the major interests of introducing exterior algebra and derivatives are Stokes theorem and Cartan magic formula. Before this however, we need to introduce what a manifold boundary is:

Definition: boundary of a submanifold.

Recall that a k -dimensional submanifold $S \subset \mathbb{R}^n$ is covered by local parametrizations $\varphi : U \rightarrow S$, with $U \subset \mathbb{R}^k$ open, exactly as introduced earlier. To allow S to have an edge, we have to consider special charts $\varphi : U \rightarrow S$ with $U \subset \mathbb{H}^k$ open and $\mathbb{H}^k$ defined by the closed half-space $\mathbb{H}^k = \{(u^1, \dots, u^k) : u^k \geq 0\}$.

A point $p \in S$ is a **boundary point** if it lies in the image of some such half-space chart with $u^k = 0$. The fact that one parameter (u^k) is forced to be zero means we cannot parametrize p with a full open ball: there has to be a coordinate "trimmed" (equal to zero) in order to parametrize it. All other points of S are **interior points**.

Theorem: boundary submanifold

The collection of all boundary points is ∂S , itself a $(k-1)$ -dimensional submanifold. S is called **boundaryless** if $\partial S = \emptyset$: every point of S admits an ordinary parametrization from an open subset of $\mathbb{R}^k$, with no half-space chart ever required.

Proof

Admitted. The reader may refer to [3] for details.

Examples:

The closed disk $D^2 = \{(x, y) : x^2 + y^2 \leq 1\} \subset \mathbb{R}^2$ needs half-space charts near the unit circle (we cannot fit the neighborhood of a point on the unit circle inside a ball: there will always be a part that does not belong to it: we need to trim, i.e. force a parameter to zero), and $\partial D^2 = S^1$.

The circle S^1 itself, parametrized locally by ordinary open intervals $\theta \in (\theta_0 - \epsilon, \theta_0 + \epsilon)$ around every point, never requires a half-space chart: $\partial S^1 = \emptyset$.

Generalized Stokes theorem

For a k -form ω and an oriented $(k+1)$ -dimensional submanifold M with boundary ∂M ,

$$\int_M d\omega = \int_{\partial M} \omega \quad (66)$$

This simple identity generalizes both the fundamental theorem of calculus ($k=0$), Green's theorem and the classical curl theorem ($k=1$), and the divergence theorem ($k=2$). The exterior derivative is defined on purpose so that this holds: $d\omega$ is the local density of ω 's flux through a boundary, and integrating that density over a region recovers the total flux through the region's boundary.

A natural consequence of Stokes theorem is the concept of closed and exact forms.

Definition: exactness and closeness of a form

An arbitrary k -form ω is said to be **closed** if $d\omega = 0$. It is said to be **exact** if it can be written as $\omega = d\alpha$ with α a $(k-1)$ -form.

If $\omega = d\alpha$ is exact and N is any compact k -manifold with $\partial N = \emptyset$, Stokes gives

$$\int_N \omega = \int_N d\alpha = \int_{\partial N} \alpha = 0. \quad (67)$$

Thus, every exact form integrates to zero over every boundaryless compact submanifold. Conversely, a non-zero integral of a form ω certifies it is not exact, even if it is closed ($d\omega = 0$).

De Rham cohomology is the systematic bookkeeping of which closed forms fail this integral test, and on which submanifolds.

Remark: Maxwell's equations in exterior form.

The condition $d\Omega = 0$ is $\nabla \cdot \vec{B} = 0$ (Gauss's law for magnetism), which says there are no magnetic monopoles (cf notes on electromagnetism).

All four Maxwell equations take elegant form with exterior derivatives. Define the electromagnetic 2-form $F = dA$ called "Faraday tensor" where A is the gauge one-form (the four-potential). Then:

$$\begin{cases} dF &= 0 & (\text{Faraday and Gauss for } B) \\ d \star F &= J & (\text{Ampère-Maxwell and Gauss for } E) \end{cases} \quad (68)$$

where $\star$ is the Hodge star (requiring a metric) and J is the current 3-form. The identity $dF = d(dA) = 0$ follows automatically from $d^2 = 0$: the homogeneous Maxwell equations are a pure consequence of exterior algebra.

Finally, before moving on to a quick introduction to Cohomology, we need to introduce Cartan's magic formula, as it is used in analytical mechanics to prove Liouville theorem. While Lie derivative itself would require further concepts before we can define it, we make the choice to take the following formula as a definition and to admit the proof of the theorem.

Definition theorem: Lie derivative (via Cartan's magic formula)

Given a vector field X , the Lie derivative $\mathcal{L}_X$ acting on differential forms (arbitrary k -forms) is defined by **Cartan's magic formula**:

$$\begin{aligned} \mathcal{L}_X : \Lambda_k &\rightarrow \Lambda_k \\ \omega &\mapsto d(\iota_X \omega) + \iota_X(d\omega) \end{aligned} \quad (69)$$

It is called "magic" because it defines an operator that involves flowing ω along the flow of X and differentiating (a genuinely analytic construction) entirely in terms of algebraic operations d and ι_X with no reference to a particular flow, a parametrization (no charts involved) or a limit. As it combines d and ι_X , the Lee derivative preserve the degree of ω . It measures the rate of change of ω as it is dragged along the flow generated by X . On functions, ($k = 0$ so $\iota_X f = 0$ trivially) the formula collapses to $\mathcal{L}_X f = \iota_X(df) = df(X) = X(f)$. The Lie derivative of function is simply its directional derivative along X .

C. Toward cohomology

The sequence of spaces and maps:

$$0 \xrightarrow{d} \Lambda_0 \xrightarrow{d} \Lambda_1 \xrightarrow{d} \Lambda_2 \xrightarrow{d} \cdots \xrightarrow{d} \Lambda_n \xrightarrow{d} 0 \quad (70)$$

is called the **de Rham complex** (no relationship with complex numbers; complex in mathematics can also mean a sequence of spaces connected by maps). The identity $d^2 = 0$ means the image of each d is contained in the kernel of the next. Every exact form is closed ($d^2 = 0$), but the converse depends on topology. This is the content of de Rham cohomology, which encodes topological information about the manifold in the failure of closed forms to be exact.

$d^2 = 0$ and the symplectic form

The nilpotency $d^2 = 0$ follows from the equality of mixed partial derivatives: applying d twice to a function f gives $d(df) = \frac{\partial^2 f}{\partial x^i \partial x^j} dx^i \wedge dx^j = 0$, because $\frac{\partial^2 f}{\partial x^i \partial x^j}$ is symmetric in i, j while $dx^i \wedge dx^j$ is antisymmetric. The contraction of a symmetric and an antisymmetric tensor is zero.

In Hamiltonian mechanics, the **tautological one-form** $\theta = p_\alpha dq^\alpha$ on the cotangent bundle $T^*\mathcal{C}$ has exterior derivative:

$$\omega = d\theta = dq^\alpha \wedge dp_\alpha \quad (71)$$

This 2-form ω is the **symplectic form**. Since $\omega = d\theta$, nilpotency immediately gives $d\omega = d^2\theta = 0$: the symplectic form is always **closed**, without any computation. It is also non-degenerate (it pairs dq^α and dp_α non-trivially). A closed non-degenerate 2-form is the definition of a symplectic structure. Its n -th exterior power $\omega^{\wedge n} = \omega \wedge \cdots \wedge \omega$ is the Liouville volume form on phase space, preserved by Hamiltonian flows.

The identity $d^2 = 0$ means that at every degree k , the image of $d : \Lambda_{k-1} \rightarrow \Lambda_k$ is contained in the kernel of $d : \Lambda_k \rightarrow \Lambda_{k+1}$:

$$\text{im}(d : \Lambda_{k-1} \rightarrow \Lambda_k) \subseteq \ker(d : \Lambda_k \rightarrow \Lambda_{k+1}) \quad (72)$$

Definition: k -th de Rham cohomology group

The **k -th de Rham cohomology group** measures the failure of this inclusion to be an equality:

$$H_{dR}^k(S) = \frac{\ker(d : \Lambda_k \rightarrow \Lambda_{k+1})}{\text{im}(d : \Lambda_{k-1} \rightarrow \Lambda_k)} = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}} \quad (73)$$

Here, the quotient set $H_{dR}^k(S)$ is an equivalence class of closed k -forms, where two closed forms are equivalent if they differ by an exact form: $\omega \sim \omega + d\alpha$. Informally, we take the set of closed form and identify forms that differs by an exact form: two closed forms ω_1 and ω_2 represent the same element in $H_{dR}^k(S)$ if $\omega_1 - \omega_2 = d\alpha$ for some $(k-1)$ -form α . If $H_{dR}^k(S) = 0$, every closed k -form on S is exact, there is no obstruction. If $H_{dR}^k(S) \neq 0$, closed forms exist that are not exact, and this failure is a property of the topology of S , not of the forms themselves.

De Rham cohomology converts a differential-analytic question (when is a closed form exact?) into a topological invariant of the space. The dimensions $b_k = \dim H_{dR}^k(S)$ are called **Betti numbers** and count independent "holes" of dimension k in S :

- b_0 : number of connected components.
- b_1 : number of independent loops that cannot be contracted to a point (one-dimensional holes).
- b_2 : number of independent closed surfaces that are not boundaries of a solid (two-dimensional holes).

Concrete example. On the circle S^1 , the one-form $\omega = d\theta$ is closed (clearly, since $d(d\theta) = 0$). Is it exact? Suppose $\omega = d\theta = df$ for some globally defined function $f : S^1 \rightarrow \mathbb{R}$. Locally this means $df/d\theta = 1$ so $f = \theta + C$. But $\theta = 0$ and $\theta = 2\pi$ represent the same point of the circle, whereas $f(0) = C$ and $f(2\pi) = 2\pi + C$: thus f cannot be well-defined on the circle. So $d\theta$ is closed but not exact: $H_{dR}^1(S^1) \neq 0$, with $b_1 = 1$. The circle has one loop. On $\mathbb{R}$ (contractible), every closed form is exact, and all cohomology groups vanish for $k \geq 1$.

Connection to physics.

- In electromagnetism, the magnetic field satisfies $\nabla \cdot \vec{B} = 0$, i.e. $dB = 0$ where B is the associated 2-form. Locally (on contractible regions) this implies $B = dA$ for a vector potential A . But on spaces with nontrivial H_{dR}^2 (such as the exterior of a magnetic monopole), a global vector potential need not exist: the Dirac monopole quantization condition is a statement about H_{dR}^2 .
- In Hamiltonian mechanics, the symplectic form ω is closed by construction ($d\omega = 0$). Whether it is exact depends on the topology of phase space. On $T^*\mathcal{C}$, it is always exact ($\omega = d\theta$), but on more general symplectic manifolds (such as compact phase spaces arising in field theory or certain reduced systems), ω may represent a nontrivial cohomology class, with physical consequences for quantization (the Bohr-Sommerfeld conditions are an early version of this).
- More generally, the **de Rham theorem** states that $H_{dR}^k(S)$ is isomorphic to the singular cohomology $H^k(S, \mathbb{R})$: a purely topological invariant defined by counting cycles and boundaries in a triangulation of S . This is an extraordinary bridge between smooth calculus and topology.

1.1.4 Abstract manifold

Everything above defined manifolds as subsets of an ambient $\mathbb{R}^{n+r}$. For most of classical and quantum mechanics, this is sufficient: configuration spaces, phase spaces, and constraint surfaces all arise naturally as submanifolds of some Euclidean space.

However, two situations force us to go further:

1. Intrinsic geometry: when one wants to define curvature, distances, or geodesics purely from within the manifold, without reference to how it sits in an ambient space. General relativity is the canonical example: spacetime is not naturally embedded in any higher-dimensional flat space.
2. Quotient constructions: some manifolds arise as equivalence classes (e.g. the space of rotations $SO(3)$, or the projective space $\mathbb{RP}^n$) and have no natural ambient embedding.

An abstract manifold is defined by specifying an atlas directly: a collection of charts with smooth transition maps, without requiring any ambient space. Whitney's embedding theorem guarantees that any smooth abstract manifold of dimension n can be embedded in $\mathbb{R}^{2n}$, so the two approaches are equivalent in principle. But the intrinsic definition is more natural for many purposes.

When we reach symplectic geometry and Hamiltonian mechanics, we will need the abstract definition to handle phase spaces and the space of rotations $SO(3)$ intrinsically.

This section will be completed later.

1.1.5 Summary

- ⊛ **The total derivative df_p .** Given $f : U \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$ and a point $p \in U$, the total derivative is the unique linear map $df_p : \mathbb{R}^m \rightarrow \mathbb{R}^k$ such that:

$$\lim_{\|h\| \rightarrow 0} \frac{\|f(p+h) - f(p) - df_p(h)\|}{\|h\|} = 0 \quad (74)$$

It is the *best linear approximation* to f near p . Indeed, the above definition is equivalent to write $f(p+h) = f(p) + df_p(h) + r(h)$ with $r(h) = o(\|h\|)$. Notation: df_p or Df_p . It is a *linear map*, not a number, not a matrix, not a vector.

- ⊛ **The Jacobian matrix $J(p)$.** Once bases are chosen for $\mathbb{R}^m$ and $\mathbb{R}^k$ (usually the standard ones), the total derivative is represented by the $k \times m$ matrix of partial derivatives:

$$J_p = \left(\frac{\partial f^i}{\partial x^j} \Big|_p \right) \quad i = 1, \dots, k, \quad j = 1, \dots, m \quad (75)$$

The Jacobian is the *coordinate matrix* of df_p . The relation is $df_p(v) = J_p v$ for $v \in \mathbb{R}^m$ written as a column vector. The Jacobian depends on the choice of bases; the total derivative does not.

- ⊛ **The differential at a point p df_p .** This name is usually only given to scalar function $f : U \subset \mathbb{R}^m \rightarrow \mathbb{R}$: we call their total derivative at p the "differential".
- ⊛ **Chain rule** If $g \circ f$ is defined for two maps f and g of total derivatives at p df_p and dg_p represented in some basis by Jacobian matrices J_p and J'_p , then $d(g \circ f)_p = dg_{f(p)} \circ df_p$, of matrix form $J'_p J_p$.
- ⊛ **Submanifold (graph description)** $S \subset \mathbb{R}^{n+r}$ is a smooth submanifold of dimension n and codimension r if every point $p \in S$ has a neighborhood U_p in which S looks like a graph: r coordinates can be expressed with the n remaining free ones (projected on U_p via $\pi : \mathbb{R}^{n+r} \rightarrow \mathbb{R}^n$) with smooth maps.

$$\exists f_p : \mathbb{R}^n \rightarrow \mathbb{R}^r / S \cap U_p = \{(x, f_p(x)), x \in \pi(U_p)\} \quad (76)$$

Submanifold (level set description): Equivalently, a submanifold can locally be described by a level set (or locus) of a smooth map $F : U_p \subset \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r$:

$$S \cap U_p = F^{-1}(\{c\}) = \{x \in U_p / F(x) = c\} \quad (77)$$

- ⊛ **Regular value theorem: a submanifold characterization** If $c \in \mathbb{R}^r$ is a regular value of $F : U \subset \mathbb{R}^{n+r} \rightarrow \mathbb{R}^r$ (i.e. if $\text{rank}[J_p] = r$ at every $p \in F^{-1}(\{c\})$), then $S = F^{-1}(\{c\})$ is a smooth n -dimensional submanifold with:

$$T_p S = \ker J(p) \quad \forall p \in S \quad (78)$$

The tangent space is the kernel of the Jacobian: the directions that fulfill all constraints. Exactly at singular points, where $\text{rank } J(p) < r$: the constraints become linearly dependent, the implicit function theorem fails, and the level set may develop a crossing, cusp, or other singularity.

- ⊛ **Implicit function theorem:** it proves the equivalence of the two descriptions of a submanifold at every regular point: it converts a level set into a graph by solving for r dependent coordinates in terms of n free ones.
- ⊛ **Tangent space** The tangent space of $\mathbb{R}^i$, $T_p\mathbb{R}^i$ is always defined, but usually unnoticed because it is canonically isomorph to $\mathbb{R}^i$. This isomorphism breaks on manifolds and submanifolds because of their geometric structure. At every regular point $p \in S$, the tangent space T_pS is an n -dimensional subspace of $T_p\mathbb{R}^{n+r}$, characterized as $\ker J(p)$. Therefore, on a manifold the total derivative at a regular point p transforms a map $F : U \subset S \rightarrow \mathbb{R}^r$ into a linear map $dF_p : T_pS \rightarrow T_p\mathbb{R}^r$. Tangent vectors in this space are often given the interpretation of a "velocity on S ", by analogy to the 1d interpretation of tangent and derivatives.
- ⊛ **Submanifold dimension** Be careful that a submanifold is not necessarily a vector space. However, its associated tangent space is. When we mention a "dimension n " submanifold, it is simply the number of independent parameters to fix all the others (graph view). However, the associated tangent space at a point will indeed have dimension n .
- ⊛ **Chart:** A pair (U_p, φ) where $U_p \subset S$ is open with $p \in U_p$ and $\varphi : U_p \rightarrow V_p \subset \mathbb{R}^n$ is a smooth bijection with smooth inverse. It assigns n real coordinates to each point of U_p , where n is the submanifold dimension. The inverse $\varphi^{-1} : V_p \rightarrow U_p$ is called the **local parametrization**. A chart is a local coordinate description of part of the manifold. Different charts describe the same geometry in different coordinates; physics cannot depend on the choice.
- ⊛ **Atlas:** A collection of charts covering all of S . Every submanifold admits one (implicit function theorem guarantees a chart near every regular point). On an abstract manifold, the atlas *is* the definition.
- ⊛ **Transition maps:** On overlapping charts, $\varphi_\beta \circ \varphi_\alpha^{-1}$ is called a transition map if it is a smooth map between open subsets of $\mathbb{R}^n$. Smoothness of transition maps is the compatibility condition ensuring the atlas is consistent.
- ⊛ **Tangent bundle** The collection of all tangent spaces of the submanifold:

$$TS = \bigsqcup_{p \in S} T_pS = \{(p, v) | p \in S, v \in T_pS\} \quad (79)$$

It is in itself a $2n$ dimensional manifold where n is the dimension of S .

- ⊛ **Vector field** A vector field on S is a smooth map: $X : S \rightarrow TS$: it assigns to each point of the manifold a tangent vector. Written with a chart φ of coordinates $(q^1, \dots, q^n)$, it takes the form:

$$X = X^i(q^1, \dots, q^n) \frac{\partial}{\partial q^i} \quad (80)$$

With $X^i : V \subset \mathbb{R}^n \rightarrow \mathbb{R}$ some smooth functions. We say that $\frac{\partial}{\partial q^i}$ are basis vectors induced by the chart: they take a point $p \in S$ and output $\frac{\partial p}{\partial q^i}$; the associated tangent vector.

- ⊛ **Vector field - alternative definition** It is also possible to define vector field using smooth functions $f : S \rightarrow \mathbb{R}$ as:

$$Xf : S \rightarrow \mathbb{R} \\ p \mapsto Xf(p) = X_p(f) = X^i(p) \frac{\partial f}{\partial q^i}(p) \quad (81)$$

In that case a vector field can be defined as a derivation; i.e. if it is a linear map on smooth maps on S and satisfies the Leibniz rule. In other words: $X : \mathcal{C}^\infty(S) \rightarrow \mathcal{C}^\infty(S)$ is linear and $X(fg) = fX(g) + gX(f)$. This definition is chart independent.

⊛ **dual space**

For a vector space V , the set of linear forms $f : V \rightarrow \mathbb{R}$ is a vector space called "dual" of V , noted V^* . Elements of V^* are often called "covectors".

- ⊛ **Cotangent space and bundle** When the considered vector space is the tangent space at $p \in T_p S$, we call its dual the cotangent space $T_p^* S$. It is an n -dimensional vector space. Similarly, the cotangent bundle denotes the collection of cotangent space over the submanifold:

$$T^* S = \bigsqcup_{p \in S} T_p^* S = \{(p, \xi) | p \in S, \xi \in T_p^* S\} \quad (82)$$

- ⊛ **Dual basis** In coordinates $(x^1, \dots, x^n)$ induced by a chart, a covector has a similar expression as the differential of a map f :

$$df_p = \frac{\partial f}{\partial x^i}(p) dx^i|_p \quad (83)$$

$dx^i|_p$ form the dual basis of $T_p^* S$; they are also linear form dual to the tangent space. In other words, $dx^i|_p : T_p S \rightarrow \mathbb{R}$ is a placeholder, a map eating a tangent vector to output a number.

⊛ **Duality**

The basis vectors of the tangent space $\partial_{x_i}|_p = \frac{\partial}{\partial x^i}|_p$ and of its dual the cotangent space $dx^i|_p$ satisfies a specific relation that represent their duality:

$$dx^i|_p(\partial_{x_j}|_p) = \delta_j^i \quad (84)$$

The index placement is intentional: we denote the coordinates of vectors v with an upper index ($v = v^i \frac{\partial}{\partial x^i}|_p$), and covector with a lower index ($\xi = \xi_i dx^i|_p$ recalling that $dx^i|_p$ is a covector but carries an upper index as the vector x^i on which an exterior derivative is applied).

- ⊛ **One forms:** they are the analogous construction of vector fields on the cotangent bundle. A cotangent vector is always defined near a point $p \in S$. A one form assigns a cotangent vector at every regular point of the submanifold:

$$\omega = \omega_i(x) dx^i \quad (85)$$

where the components $\omega_i : V \subset \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth. Evaluated at p ; dx^i is the dual basis at $T_p S$ noted $dx^i|_p$.

- ⊛ **Riemannian metric:** A smooth assignment $g_p : T_p S \times T_p S \rightarrow \mathbb{R}$ of an inner product at each point symmetric, positive and definite. In coordinates with a chart, $g_p(u, v)$ is represented by a smooth matrix $g_{\alpha\beta}(x) = g_p(\frac{\partial}{\partial x^\alpha}, \frac{\partial}{\partial x^\beta})$, symmetric positive definite. Provided two vectors $u = u^\alpha \frac{\partial}{\partial x^\alpha}$ and $v = v^\beta \frac{\partial}{\partial x^\beta}$, we get:

$$g_p(u, v) = g_{\alpha\beta}(x) u^\alpha v^\beta \quad (86)$$

A submanifold of a Euclidian ambient space naturally inherit its metric : if $u, v \in T_p S$, we can define $g_p(u, v) = u \cdot v$ (dot product). This metric is the "induced metric", or "first fundamental form".

⊛ **Properties of a metric**

- Lengths on a curve $\gamma : [a, b] \rightarrow S$: $l(\gamma) = \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt = \int_a^b \sqrt{g_{\alpha\beta} \dot{\gamma}^\alpha \dot{\gamma}^\beta} dt$.
- Angle between vectors: $\forall u, v \in T_p S, \cos(\theta) = \frac{g_p(u, v)}{\|u\| \|v\|}$.
- Geodesics: shortest length between two manifold point, satisfying $\ddot{\gamma}^\alpha + \Gamma_{\beta\nu}^\alpha \dot{\gamma}^\beta \dot{\gamma}^\nu = 0$.

⊛ **musical isomorphisms**

The maps flat $\flat : T_p S \rightarrow T_p^* S$ and sharp $\sharp : T_p^* S \rightarrow T_p S$ defined by $\flat(v) = g_p(v, \cdot)$ and its inverse $\sharp(\xi) = g^{\alpha\beta} \xi_\beta \partial_\alpha$ allow for a canonical identification between a vector space and its dual. They require a metric; but not necessarily a chart to exist: this is the meaning of the word "canonical"; coordinate independent.

⊛ **Inner product (reminder)**: an inner product is simply a map asking for two vectors and returning a scalar. The typical example is the dot product for Euclidian spaces: $\cdot : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}; u \cdot v = \sum_i u^i v^i$

⊛ **Tensor product** we can define a tensor product $\otimes$ as the product of two vector spaces built in a way that we remove (or force to be equal) any pairs of vectors representing the same linear combination. The tensor product of two basis a and b of vector spaces E and F of dimension n and m form a basis of the vector space $E \otimes F$ of dimension nm .

⊛ **Tensor product space** A tensor product can be performed on a vector space (contravariant tensor) or its dual (covariant tensor). We call $T_k(V) = \bigotimes_{i=1}^k V^*$ the covariant tensor space of degree k , and $T^k(V) = \bigotimes_{i=1}^k V$ its dual, the contravariant tensor space of degree k . Finally, a tensor built on r copies of a given vector space V and s of its dual V^* generally belongs to the mixed tensor space: $T_s^r(V) = T^r(V) \otimes T_s(V)$.

⊛ **Tensor algebra** The direct sum over all possible degrees of a tensor product space is called the tensor algebra:

$$T(V) = \bigoplus_{k=0}^{\infty} T^k(V) \quad (87)$$

With the convention that $T^0(V) = \mathbb{K}$ (the associated scalar field). Note that a similar tensor algebra can of course be built for covariant tensors (still assuming $T_0(V) = \mathbb{K}$).

⊛ **Exterior (or wedge) product** For two vectors (or alternatively two one forms) α and β , the exterior product (or wedge product) $\wedge$ antisymmetrizes the tensor product. It is defined by:

$$\alpha \wedge \beta = \binom{p+q}{q} \text{Alt}(\alpha \otimes \beta) \quad (88)$$

With the Alt (alternate) operator an antisymmetrizing operator:

$$\text{Alt}(v_1 \otimes \cdots \otimes v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} \text{sgn}(\sigma) (v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(k)}) \quad (89)$$

The result of the wedge product of two vector is a bi-vector. The wedge product of two one-forms is a two-form. Note that for a p-vector or p-form α and a q-vector or q-form β , $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$ and $\alpha \wedge \alpha = 0$. The wedge product basically measures the oriented area, volumes... between vectors.

- ⊛ **Exterior product space and algebra** The set of all k -vector built from a vector space V (or k -form) is noted $\Lambda^k(V)$ (or $\Lambda_k(V)$). It is in itself a vector space of dimension $\binom{n}{k}$ with $n = \dim(V)$. It is a subset of the tensor product space $T^k(V)$ (respectively $T_k(V)$). Note that $\Lambda^k(V) = \{0\}$ for $k > n$. The exterior algebra thus:

$$\Lambda(V) = \bigoplus_{k=0}^{\infty} \Lambda^k(V) = \bigoplus_{k=0}^n \Lambda^k(V) \quad (90)$$

Note that the unique n -form of $\Lambda_n(V)$ is called "volume form".

- ⊛ **Relation to cross product** It turns out that in three dimension, $\dim(\Lambda^2(\mathbb{R}^3)) = \binom{3}{2} = 3$. Hence, bivectors and vectors has the same dimension. Using the Euclidian metric and an operator called "Hodge star" $\star : \Lambda_2(\mathbb{R}^3) \rightarrow \Lambda_1(\mathbb{R}^3)$, an identification is possible. But it turns out more of a curse, as many uses cross product without understanding the difference between a bi-vector and a vector. The magnetic field itself is technically a two form.
- ⊛ **Interior product** The interior product ι_X is the opposite operation to the exterior product; lowering the degree of a form or a vector. Given a vector field X and a k -form $\omega \in \Lambda_k(V)$,

$$\iota_X(\omega) = (X, v_1, \dots, v_{k-1}) = \omega(X, v_1, \dots, v_{k-1}) \in \Lambda_{k-1}(V) \quad (91)$$

The interior product basically uses a vector field and plug it in one of the slots of a k -form to output a $(k-1)$ -form. For a k -vector, it will conversely use a one form α :

$$\iota_\alpha(v_1 \wedge \dots \wedge v_k) = \sum_{i=1}^k (-1)^{i-1} \alpha(v_i) v_1 \wedge \dots \wedge v_{i-1} \wedge v_{i+1} \wedge \dots \wedge v_k \in \Lambda^{k-1}(V) \quad (92)$$

- ⊛ **Exterior derivative**

The exterior derivative is really the one operation hidden behind many physical equations. It is the unique family of linear maps $d : \Lambda_k(V) \rightarrow \Lambda_{k+1}(V)$ such that:

1. On functions: df is the differential.
2. Linearity: $\forall \alpha, \beta \in \Lambda_k(V), d(\alpha + \beta) = d\alpha + d\beta$.
3. Graded Leibniz rule: For α a k -form and β any multilinear form, $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$.
4. Nilpotency: $d \circ d = 0$ so $\forall \omega \in \Lambda_k(V), d^2 \omega = 0$.

This definition is chart and metric free. Should we provide a chart though, the definition of the exterior derivative on a general k -form ω would be:

$$d\omega = \frac{1}{k!} \sum_{i_1, \dots, i_k} \frac{\partial \omega_{i_1, \dots, i_k}}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k} \quad (93)$$

Exterior derivative are really the operation allowing to define a gradient (on function), a curl (on one forms), a divergence (on two forms)...

- ⊛ **Boundary of a submanifold** The boundary of a submanifold S is noted ∂S and is the set of all points lying in the image of a half-space chart with one coordinate vanishing. Written with words, this means those points cannot have a neighborhood described by an open ball: there must be one coordinate vanishing somewhere. See Fig. 10 for a visual example. Note that the boundary of a boundary is always empty. It is also perfectly possible to consider a boundaryless manifold, such as a sphere or a torus.
- ⊛ **Generalized Stokes theorem** Stokes theorem is a simple yet very powerful theorem generalizing fundamental theorem of calculus and Green-Ostrogradski theorem. It states that for a k -form and an oriented $(k+1)$ -dimensional submanifold S of boundary ∂S , then

$$\int_S d\omega = \int_{\partial S} \omega \quad (94)$$

Of course, the variable of integration is still contained in the coordinate representation of ω , provided a chart. For an exact form ω (i.e. that can be written as $\omega = d\alpha$), if the submanifold is boundaryless, we naturally get a vanishing integral. Thus, on a boundaryless submanifold, Stokes theorem is a way of measuring exactness of forms.

- ⊛ **Cartan's magic formula**

Cartan's magic formula is named this way because it links differentiation and flows purely in terms of the algebraic operations d and ι_X . We use it to provide here a first definition to Lie derivatives $\mathcal{L}_x$:

$$\begin{aligned} \Lambda_k(V) &\rightarrow \Lambda_k(V) \\ \omega &\mapsto d(\iota_X \omega) + \iota_X(d\omega) \end{aligned} \quad (95)$$

A Lie derivative is a measure of how ω change as it is moving through the flow generated by the vector field X . It preserves the degree of the k -form ω .

- ⊛ **De Rham complex**

The De Rham complex is the name of the sequence of space connected by the exterior derivative:

$$0 \xrightarrow{d} \Lambda_0 \xrightarrow{d} \Lambda_1 \xrightarrow{d} \Lambda_2 \xrightarrow{d} \cdots \xrightarrow{d} \Lambda_n \xrightarrow{d} 0 \quad (96)$$

As d is nilpotent, the image of each $d : \Lambda_{k-1} \rightarrow \Lambda_k$ is included in the kernel of the next $d : \Lambda_k \rightarrow \Lambda_{k+1}$. De Rham cohomology is a way of measuring the consequence of topology in the manifold.

- ⊛ **k-th De Rham cohomology group**

The k -th de Rham cohomology group is a way of measuring how much closed forms fails to be exact

$$H_{dR}^k(S) = \frac{\ker(d : \Lambda_k \rightarrow \Lambda_{k+1})}{\operatorname{im}(d : \Lambda_{k-1} \rightarrow \Lambda_k)} \quad (97)$$

The dimension of each k -th de Rham cohomology group is called Betti number and counts the number of independent holes/cycles of dimension k in the manifold. Cohomology seems abstract at first but is a way of knowing a lot of information only from the geometry [4]. We will get back to it on our chapter about topology.

References

- [1] Truman Botts. Mathematics: *calculus on manifolds*. by michael spivak. benjamin, new york, 1965. 158 pp. paper, \$2.95; cloth, \$7. *Science*, 153(3732):164–165, 1966. doi: 10.1126/science.153.3732.164.b. URL <https://www.science.org/doi/abs/10.1126/science.153.3732.164.b>.
- [2] Theodore Frankel. *The Geometry of Physics: An Introduction, Second Edition*. Cambridge, 2003. ISBN 0521539277.
- [3] Victor Guillemin and Alan Pollack. *Differential Topology*. Prentice-Hall, Englewood Cliffs, NJ, 1974. ISBN 0-13-212605-2.
- [4] Mikio Nakahara. *Geometry, Topology and Physics*. CRC Press, 2 edition, October 2018. ISBN 9781315275826. doi: 10.1201/9781315275826. URL <https://www.taylorfrancis.com/books/9781420056945>.
- [5] W. Rudin. *Principles of Mathematical Analysis*. International series in pure and applied mathematics. McGraw-Hill, 1976. ISBN 9780070856134. URL <https://books.google.fr/books?id=kwqzPAAACAAJ>.